

Phase 2A — DB-driven Theming (Batch 1)

A Super-Admin-managed theming system with **three independent scopes**, each fully CRUD-able and applied **live via CSS variables** — no redeploy is needed to switch or edit a theme.

Scopes

Scope	Surface	Applied on
website	Public marketing site	/ and public pages
splash	Intro / module-selection screen	/modules
app	Internal authenticated portals	/admin and /app

Tenant public sites (/site/[slug]) intentionally keep the baked-in baseline.

Data model

themes table (migration a1b2c3d4e5f6_phase2a_themes):

column	type	notes
id	int PK	
scope	string	website splash app (indexed)
name	string	
is_default	bool	exactly one default per scope (service-enforced)
tokens	JSON	full design-token blob
created_at / updated_at	timestampz	

The token blob is flexible JSON so Batch 2 (per-section content editing) can layer on top without a schema change. `theme_service.effective_tokens()` merges each theme's stored tokens over `DEFAULT_TOKENS`, so new token keys get sane defaults on existing rows automatically.

Token keys

Colors: `primary`, `primary_hover`, `primary_contrast`, `secondary`, `secondary_hover`, `secondary_contrast`, `accent`, `background`, `page_bg`,

`surface_muted`, `text`, `text_muted`, `border`, `success`, `warning`, `error`.
 Typography: `font_heading`, `font_body`, `font_size_base`, `heading_weight`,
`body_weight`. Shape/density: `radius_base`, `button_radius`, `button_style`
 (`solid` | `outline`), `button_text_transform` (`none` | `uppercase`), `density`,
`shadow_level`. Also `base_mode` (`light` | `dark`), `section_backgrounds`
 (Batch 2 hook), `logo_url`.

Seeded themes (idempotent)

Four themes are seeded **per scope** (12 rows total) by
`theme_service.seed_themes()`, called from `app.db.seed`:

1. **BaseCenter Default** — the default; FreshBooks-style (Science Blue `#0075DD`, white surfaces, near-black text, Inter, lightly rounded solid buttons).
2. **Light** — cooler slate/blue palette.
3. **Dark** — true dark (`base_mode`: `dark`, dark surfaces, light text).
4. **Bold Contrast** — dramatic (violet `#7C3AED`, pink accent, Montserrat headings, sharp 4px corners, uppercase buttons).

Seeding never clobbers edited tokens (upsert by `(scope, name)`), and always guarantees exactly one default per scope.

API

Public (no auth)

- GET `/api/v1/themes/active?scope=<scope>` → `{ scope, tokens }` — the effective tokens of the scope's default theme. Falls back to the baked-in baseline for unknown/empty scopes (never 500s).

Super Admin (`get_current_superuser`)

- GET `/api/v1/admin/themes?scope=<scope>` — list themes in a scope.
- POST `/api/v1/admin/themes` — create.
- GET `/api/v1/admin/themes/{id}` — fetch one.
- PUT `/api/v1/admin/themes/{id}` — update name/tokens.
- DELETE `/api/v1/admin/themes/{id}` — delete (blocked for the default).
- POST `/api/v1/admin/themes/{id}/duplicate` — copy (never default).
- POST `/api/v1/admin/themes/{id}/set-default` — make sole default for scope.

All writes are audited via `audit_service` (`theme.create`, `theme.update`,
`theme.delete`, `theme.duplicate`, `theme.set_default`).

Live application mechanism

`globals.css` declares the baseline as `:root` CSS custom properties, and
 Tailwind v4's `@theme inline` compiles every utility to read those variables.
 The client `ThemeManager` (mounted globally in `app/layout.tsx`):

1. Resolves the scope from the current path (`lib/theme.ts` → `scopeForPath`).
2. Fetches `GET /themes/active?scope=...` (cached per scope in memory).

3. Applies the tokens as CSS variables on `<html>` (`lib/theme.ts` → `applyTokens`), including primary/secondary tint & shade derivation, radius scaling, and — for dark themes — remapping the Tailwind `neutral-*` scale so gray text/surfaces invert.

Because utilities reference these variables, changing the default theme (or editing tokens) is reflected on the **next page load** of any surface in that scope, with **no rebuild**. If the fetch fails, the baked-in FreshBooks Default baseline remains in effect (graceful fallback).

SA “Appearance” UI

At `/admin/appearance` (linked in the Super Admin sidebar). Three tabs — Public Website / Intro-Splash / Internal App — each a full theme manager: theme list with default badge, color pickers, font/typography controls, radius/density/button-style/shadow controls, light-dark toggle, logo URL, a live preview panel, plus Duplicate, Delete (guards the default), Set-as-Default, and New. The **Internal App** tab additionally embeds the sidebar navigation label editor (same `/admin/settings/sidebar` endpoints), keeping internal theme and nav labels in one place.

`/modules` splash

Rebuilt as a responsive grid of **all modules** fetched from `GET /api/v1/modules` (with a baked-in fallback catalog). Paid modules show `$5/mo` ; the Website Builder is badged **Free • Included**. Selecting a card reveals a sticky CTA flowing into the existing signup path (`/register` → `/app/signup`). The splash-scope theme is applied automatically by `ThemeManager` .